

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2024 Worked Solutions

Oct 2024 paper rebuilt with the worked solution after each question.

9

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2024

Oct 2024 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Straight Line

1. The line l_1 passes through the point $A(-5, 20)$ and the point $B(3, -4)$.

- (a) Find an equation for l_1 giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB

- (b) Find an equation for l_2 giving your answer in the form $px + qy + r = 0$, where p , q and r are integers.

(3)

(Total for Question 1 is 6 marks)

Worked Solution - Question 1

1. Find the gradient of l_1

$$m = \frac{-4 - 20}{3 - (-5)} = \frac{-24}{8} = -3.$$

2. Find the equation of l_1

Using $B(3, -4)$, $y = -3x + c$. Then $-4 = -9 + c$, so $c = 5$.

3. Find the midpoint of AB

$$M = \left(\frac{-5 + 3}{2}, \frac{20 + (-4)}{2} \right) = (-1, 8).$$

4. Use perpendicular gradient

Since l_1 has gradient -3 , l_2 has gradient $\frac{1}{3}$.

5. Find the equation of l_2

$$y - 8 = \frac{1}{3}(x + 1), \text{ so } 3y - 24 = x + 1 \text{ and } x - 3y + 25 = 0.$$

Final answer

(a) $y = -3x + 5$.

(b) $x - 3y + 25 = 0$.

Question 2

Indices & Surds

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4} \quad (3)$$

(ii) Find the exact value of a such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form $p\sqrt{3} + q$ where p and q are fully simplified rational constants.

(4)

(Total for Question 2 is 7 marks)

Worked Solution - Question 2

1. Simplify the power bracket

$$(2x^4)^3 = 8x^{12}.$$

2. Simplify the algebraic fraction

$$\frac{3y^3(2x^4)^3}{4x^2y^4} = \frac{3y^3(8x^{12})}{4x^2y^4} = \frac{6x^{10}}{y}.$$

3. Rationalise the left side

$$\frac{16}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 8\sqrt{3} - 8.$$

4. Equate the surd parts

$$8\sqrt{3} - 8 = a\sqrt{27} + 4 = 3a\sqrt{3} + 4.$$

5. Solve for a

$$3a\sqrt{3} = 8\sqrt{3} - 12, \text{ so } a = \frac{8\sqrt{3} - 12}{3\sqrt{3}} = \frac{8}{3} - \frac{4}{3}\sqrt{3}.$$

Final answer

$$(i) \frac{6x^{10}}{y}. \quad (ii) a = -\frac{4}{3}\sqrt{3} + \frac{8}{3}.$$

Question 3

Integration

3.

In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find $\int f(x) \, dx$

(4)

(b) (i) Show that when $f'(x) = 0$

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of x for which

$$f'(x) = 0$$

(5)

(Total for Question 3 is 9 marks)

Worked Solution - Question 3

1. Rewrite $f(x)$

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} = \frac{x^2 + 10x + 25}{x^{1/2}} = x^{3/2} + 10x^{1/2} + 25x^{-1/2}.$$

2. Integrate

$$\int f(x) dx = \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c.$$

3. Differentiate $f(x)$

$$f'(x) = \frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2}.$$

4. Set f' prime equal to zero

$$\frac{3}{2}x^{1/2} + 5x^{-1/2} - \frac{25}{2}x^{-3/2} = 0.$$

5. Clear powers of x

Multiplying by $2x^{3/2}$ gives $3x^2 + 10x - 25 = 0$.

6. Use x greater than zero

$$3x^2 + 10x - 25 = (3x - 5)(x + 5), \text{ so } x = \frac{5}{3} \text{ or } x = -5. \text{ Since } x > 0, x = \frac{5}{3}.$$

Final answer

$$(a) \frac{2}{5}x^{5/2} + \frac{20}{3}x^{3/2} + 50x^{1/2} + c.$$

$$(b) x = \frac{5}{3}.$$

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10 marks

Question 4

Polynomials

4.

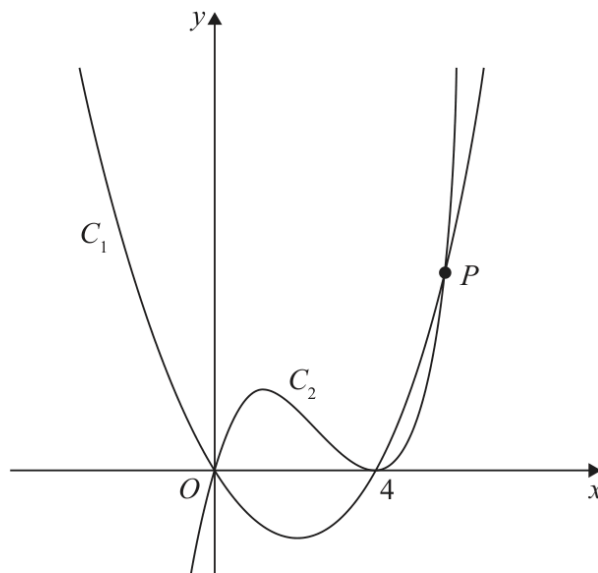


Figure 1

Figure 1 shows a sketch of part of the curves C_1 and C_2

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a quadratic function
- cuts the x -axis at the origin and at $x = 4$
- has a minimum turning point at $(2, -4.8)$

(a) find $f(x)$

(3)

Given that C_2

- has equation $y = g(x)$ where $g(x)$ is a cubic function
- cuts the x -axis at the origin and meets the x -axis at $x = 4$
- passes through the point $(6, 7.2)$

(b) find $g(x)$

(3)

The curves C_1 and C_2 meet in the first quadrant at the point P , shown in Figure 1.

(c) Use algebra to find the coordinates of P .

(4)

(Total for Question 4 is 10 marks)

Worked Solution - Question 4

1. Use the roots of C_1

Since C_1 cuts the x-axis at 0 and 4, let $f(x) = ax(x - 4)$.

2. Use the minimum point of C_1

At $x = 2$, $f(2) = -4a = -4.8 = -\frac{24}{5}$, so $a = \frac{6}{5}$.

3. Write $f(x)$

$$f(x) = \frac{6}{5}x(x - 4).$$

4. Use the roots of C_2

The cubic cuts the x-axis at the origin and meets the x-axis at $x = 4$, so $x = 4$ is a repeated root. Let $g(x) = bx(x - 4)^2$.

5. Use the point (6, 7.2)

$$7.2 = b(6)(2^2) = 24b, \text{ so } b = 0.3 = \frac{3}{10}.$$

6. Equate the curves

$$\frac{6}{5}x(x - 4) = \frac{3}{10}x(x - 4)^2.$$

7. Find the first-quadrant intersection

Ignoring the shared roots on the x-axis, divide by $x(x - 4)$ to get

$$\frac{6}{5} = \frac{3}{10}(x - 4). \text{ Hence } x = 8.$$

8. Find y

$$y = f(8) = \frac{6}{5}(8)(4) = \frac{192}{5}.$$

Final answer

$$(a) f(x) = \frac{6}{5}x(x - 4).$$

$$(b) g(x) = \frac{3}{10}x(x - 4)^2.$$

$$(c) P = (8, \frac{192}{5}).$$

Question 5

Radians

5. A plot of land OAB is in the shape of a sector of a circle with centre O .

Given

- $OA = OB = 5 \text{ km}$
- angle $AOB = 1.2 \text{ radians}$

- (a) find the perimeter of the plot of land.

(2)

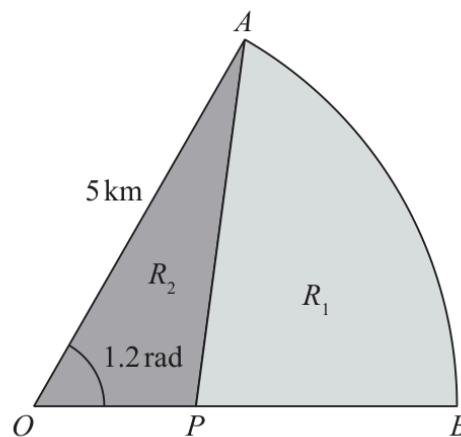


Diagram **NO**
accurately dr

Figure 2

A point P lies on OB such that the line AP divides the plot of land into two regions R_1 and R_2 as shown in Figure 2.

Given that

$$\text{area of } R_1 = 3 \times \text{area of } R_2$$

- (b) show that the area of $R_2 = 3.75 \text{ km}^2$

(3)

- (c) Find the length of AP , giving your answer to the nearest 100 m.

(4)

(Total for Question 5 is 9 marks)

Worked Solution - Question 5

1. Find the arc length

$$\text{arc } AB = r\theta = 5(1.2) = 6 \text{ km.}$$

2. Find the perimeter

$$OA + OB + \text{arc } AB = 5 + 5 + 6 = 16 \text{ km.}$$

3. Find the total sector area

$$\text{Area of } OAB = \frac{1}{2}(5^2)(1.2) = 15 \text{ km}^2.$$

4. Use the area ratio

Since $\text{area } R_1 = 3 \times \text{area } R_2$, the total area is $4 \times \text{area } R_2$.

5. Find area R2

$$\text{area } R_2 = \frac{15}{4} = 3.75 \text{ km}^2.$$

6. Find OP from triangle OAP

$$\frac{1}{2}(5)(OP) \sin 1.2 = 3.75, \text{ so } OP \approx 1.609 \text{ km.}$$

7. Use the cosine rule

$$AP^2 = 5^2 + OP^2 - 2(5)(OP) \cos 1.2, \text{ so } AP \approx 4.665 \text{ km.}$$

8. Round to the nearest 100 m

$$AP = 4.7 \text{ km to the nearest } 0.1 \text{ km.}$$

Final answer

(a) 16 km.

(b) 3.75 km^2 .

(c) $AP = 4.7 \text{ km}$.

Question 6

Graphs of Functions

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2-x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

Topic group

1. Identify the graph features

$y = \frac{1}{2-x}$ has vertical asymptote $x = 2$ and horizontal asymptote $y = 0$.

2. Find the y-intercept

At $x = 0$, $y = \frac{1}{2}$, so the y-intercept is $(0, \frac{1}{2})$.

3. Set up the intersection with the line

If $y = kx - 4$ cuts C , then $kx - 4 = \frac{1}{2-x}$.

4. Form the quadratic

$(kx - 4)(2 - x) = 1$, so $kx^2 - (2k + 4)x + 9 = 0$.

5. Use the discriminant

For at least one intersection, the discriminant is non-negative.

6. Derive the given inequality

$(2k + 4)^2 - 4(k)(9) \geq 0$, so $4(k^2 - 5k + 4) \geq 0$, hence $k^2 - 5k + 4 \geq 0$.

7. Solve the inequality

$k^2 - 5k + 4 = (k - 1)(k - 4)$, so $k \leq 1$ or $k \geq 4$.

Final answer

(a) vertical asymptote $x = 2$, y-intercept $(0, \frac{1}{2})$.

(b) $k \leq 1$ or $k \geq 4$.

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8 marks

Question 7

Trigonometric Functions

7.

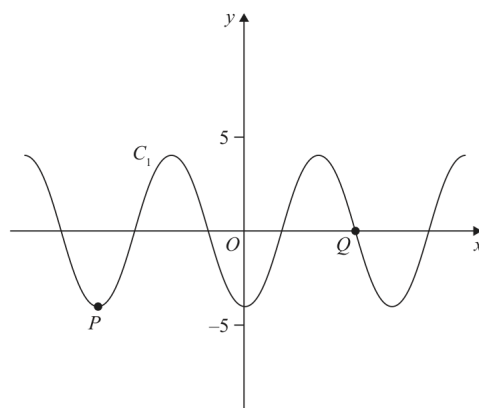


Figure 3

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4 \cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

- (i) the coordinates of P
- (ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4 \cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

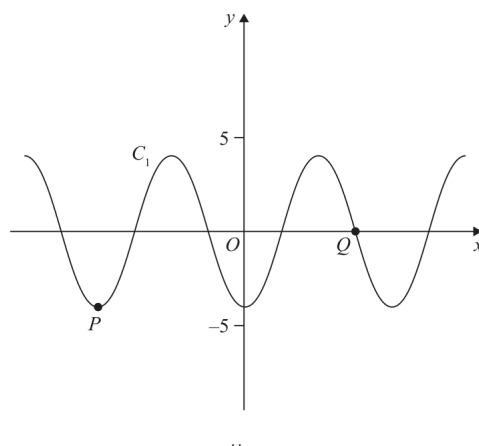
- (ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi} x$$



(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Use y equals negative $4 \cos x$

For $y = -4 \cos x$, minimum points occur at $x = 2\pi n$ with value -4 .

2. Read P

The displayed minimum to the left of the y -axis is $P = (-2\pi, -4)$.

3. Read Q

The point Q is the positive x -intercept after the maximum at $x = \pi$, so
 $Q = (\frac{3\pi}{2}, 0)$.

4. Find k for C2

For $y = -4 \cos x + k$, the maximum value is $4 + k$. Given the maximum is 11 ,
 $k = 7$.

5. Find the smallest positive minimum on C2

The minimum value is $-4 + 7 = 3$. The first positive minimum is at $x = 2\pi$, so
the point is $(2\pi, 3)$.

6. Interpret the equation graphically

$-4 \cos x = 5 - \frac{10}{\pi}x$ is the intersection of the cosine curve with the straight line
 $y = 5 - \frac{10}{\pi}x$.

7. Count intersections

The line passes through $(0, 5)$ and meets the displayed curve once, at the x -axis
point, so there is 1 solution.

Final answer

$(a)(i) P = (-2\pi, -4).$ $(a)(ii) Q = (\frac{3\pi}{2}, 0).$

$(b)(i) k = 7.$

$(b)(ii) (2\pi, 3).$

(c) 1 solution.

Question 8

Integration

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $f(x)$.

(4)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Find the y-coordinate of P

The normal is $y = -\frac{1}{24}x + 5$. At $x = 3$, $y = -\frac{3}{24} + 5 = \frac{39}{8}$.

2. Use the normal gradient

The normal gradient is $-\frac{1}{24}$, so the tangent gradient is **24**.

3. Use f prime at x equals 3

$$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k.$$

4. Find k

$$39 + 3k = 24, \text{ so } k = -5.$$

5. Integrate f prime

With $k = -5$, $f'(x) = 4x^2 - 5x + 3$, so $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c$.

6. Use point P

$$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c.$$

7. Find c

$$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}.$$

Final answer

$$k = -5 \text{ and } f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}.$$

Question 9

Differentiation

9.

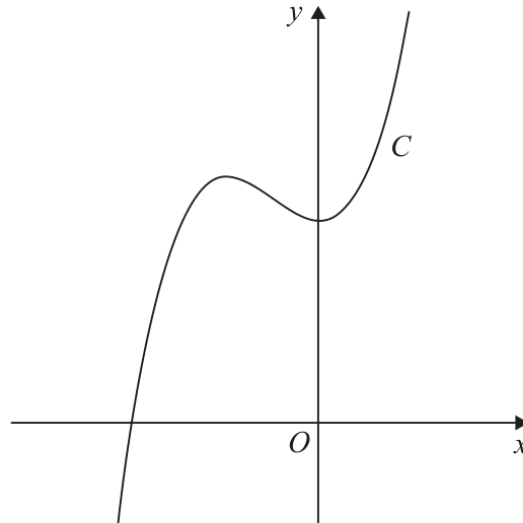


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

- (a) Deduce the range of values of x for which $f(x) \geq 0$ (1)
- (b) Find $f'(x)$ giving your answer in simplest form. (3)

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

- (c) find the x coordinate of P . (4)
- (d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,
- (i) $y = f(x - 3)$
- (ii) $y = 4f(x)$ (2)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

1. Use the positive quadratic factor

$3x^2 - 4x + 20$ has discriminant $(-4)^2 - 4(3)(20) < 0$ and positive leading coefficient, so it is always positive.

2. Deduce the sign of $f(x)$

The sign of $f(x) = (x + 5)(3x^2 - 4x + 20)$ is therefore the sign of $x + 5$.
Hence $f(x) \geq 0$ for $x \geq -5$.

3. Expand $f(x)$

$$f(x) = 3x^3 + 11x^2 + 100.$$

4. Differentiate

$$f'(x) = 9x^2 + 22x.$$

5. Find the gradient at R

$$\text{At } R(-4, 84), f'(-4) = 9(16) + 22(-4) = 56.$$

6. Use parallel tangents

For the tangent at P to be parallel, $f'(x) = 56$.

7. Solve for x

$$9x^2 + 22x - 56 = 0, \text{ so } x = \frac{-22 \pm 50}{18}. \text{ The roots are } x = -4 \text{ and } x = \frac{14}{9}.$$

8. Choose the point P

$$x = -4 \text{ is the point } R, \text{ so } x_P = \frac{14}{9}.$$

9. Transform R under y equals f of x minus 3

$y = f(x - 3)$ translates the curve 3 units to the right, so $(-4, 84)$ maps to $(-1, 84)$.

10. Transform R under y equals 4f(x)

$y = 4f(x)$ stretches vertically by scale factor 4, so $(-4, 84)$ maps to $(-4, 336)$.

Final answer

(a) $x \geq -5$.

(b) $f'(x) = 9x^2 + 22x$.

(c) $x_P = \frac{14}{9}$.

(d)(i) $(-1, 84)$.

(d)(ii) $(-4, 336)$.